

A. Manual patch for Ubuntu VM

Step 1:

Add Below Role in IAM in EC2 service.

IAM Role --> AmazonSSMManagedInstanceCore

Step2:

To start SSM service and check if running or not, check below command.

```
sudo snap install amazon-ssm-agent --classic
sudo systemctl enable snap.amazon-ssm-agent.amazon-ssm-agent
sudo systemctl start snap.amazon-ssm-agent.amazon-ssm-agent
sudo systemctl status snap.amazon-ssm-agent.amazon-ssm-agent
```

```
ubuntu@ip-172-31-14-235:~$ sudo systemctl status amazon-ssm-agent
Unit amazon-ssm-agent.service could not be found.
ubuntu@ip-172-31-14-235:~$ sudo snap install amazon-ssm-agent --classic
snap "amazon-ssm-agent" is already installed, see 'snap help refresh'
ubuntu@ip-172-31-14-235:~$ sudo systemctl enable snap.amazon-ssm-agent.amazon-ssm-agent
ubuntu@ip-172-31-14-235:~$ sudo systemctl start snap.amazon-ssm-agent.amazon-ssm-agent
ubuntu@ip-172-31-14-235:~$ sudo systemctl status amazon-ssm-agent
Unit amazon-ssm-agent.service could not be found.
ubuntu@ip-172-31-14-235:~$ sudo systemctl status snap.amazon-ssm-agent.amazon-ssm-agent
● snap.amazon-ssm-agent.amazon-ssm-agent.service - Service for snap application amazon-ssm-agent.amazon-ssm-agent
   Loaded: loaded (/etc/systemd/system/snap.amazon-ssm-agent.amazon-ssm-agent.service; enabled; vendor preset: enabled)
   Active: active (running) since Fri 2026-01-09 09:11:22 UTC; 17min ago
     Main PID: 1094 (amazon-ssm-agent)
        Tasks: 9 (limit: 1073)
       Memory: 26.6M
          CPU: 908ms
      CGroup: /system.slice/snap.amazon-ssm-agent.amazon-ssm-agent.service
              └─1094 /snap/amazon-ssm-agent/11797/amazon-ssm-agent

Jan 09 09:11:24 ip-172-31-14-235 amazon-ssm-agent.amazon-ssm-agent[1094]: 2026-01-09 09:11:23.5508 INFO [EC2Identity] EC2 registration was successful.
Jan 09 09:11:24 ip-172-31-14-235 amazon-ssm-agent.amazon-ssm-agent[1094]: 2026-01-09 09:11:23.5508 INFO [amazon-ssm-agent] Registration attempted. Resuming co
re agent startup.
Jan 09 09:11:25 ip-172-31-14-235 amazon-ssm-agent.amazon-ssm-agent[1094]: 2026-01-09 09:11:23.5509 INFO [CredentialRefresher] credentialRefresher has started
Jan 09 09:11:25 ip-172-31-14-235 amazon-ssm-agent.amazon-ssm-agent[1094]: 2026-01-09 09:11:23.5509 INFO [CredentialRefresher] Starting credentials refresher l
oop
Jan 09 09:11:25 ip-172-31-14-235 amazon-ssm-agent.amazon-ssm-agent[1094]: 2026-01-09 09:11:23.5538 WARN EC2RoleProvider Failed to connect to Systems Manager w
ith instance profile
Jan 09 09:11:25 ip-172-31-14-235 amazon-ssm-agent.amazon-ssm-agent[1094]: 2026-01-09 09:11:23.6127 ERROR EC2RoleProvider Failed to connect to Systems Manager
with SSM role creden
Jan 09 09:11:25 ip-172-31-14-235 amazon-ssm-agent.amazon-ssm-agent[1094]: status code: 400, request id: 1a68d55e-162b-46d6-8fd4-a2b30cbef746
Jan 09 09:11:25 ip-172-31-14-235 amazon-ssm-agent.amazon-ssm-agent[1094]: 2026-01-09 09:11:23.6128 ERROR [CredentialRefresher] Retrieve credentials produced e
rror: no valid creden
Jan 09 09:11:25 ip-172-31-14-235 amazon-ssm-agent.amazon-ssm-agent[1094]: status code: 400, request id: 1a68d55e-162b-46d6-8fd4-a2b30cbef746
Jan 09 09:11:25 ip-172-31-14-235 amazon-ssm-agent.amazon-ssm-agent[1094]: 2026-01-09 09:11:23.6128 INFO [CredentialRefresher] Sleeping for 28m50s before retry
ing retrieve creden
ubuntu@ip-172-31-14-235:~$
```

Step 3:

As the SSM agent is Running inside VM, check if its Reflecting in AWS Service as well in below options

1. Check at Fleet Manager in Systems Manager Service
2. In Session Manager start a new session of that VM. If terminal opens properly and its good.

Fleet Manager Info

Settings Account management

You may have unmanaged Amazon EC2 instances

You can automatically configure Amazon EC2 instances as managed instances in your current account and Region by enabling Default Host Management Configuration. [Learn more](#)

Configure Default Host Management

Managed Nodes (1)

Report Node actions

Filter

Last fetched at: 3:46 PM

	Node ID	Node s...	Name	Platfor...	Operat...	Resour...	Source ID	Ping st...	Agent ...	Ima
☐	i-0f154d67d5a0cf886	Running	Ubuntu22.04	Linux	Ubuntu	EC2 insta...	-	Online	3.3.2299.0	ami

AWS Systems Manager

Review node insights

Explore nodes

Diagnose and remediate

Just-in-time node access [New](#)

Settings

▼ Node Tools

Compliance

Distributor

Fleet Manager

Hybrid Activations

Inventory

Patch Manager

Run Command

Session Manager

State Manager

▼ Change Management Tools

Automation

AWS Systems Manager

Session Manager

Start a session

Step 1

Specify target

Step 2 - optional

Specify session document

Step 3

Review and launch

Specify target

Select an instance to connect to using Session Manager.

Reason

Reason for session – optional

The reason for connecting to the instance. This value is included in the details of the event created by AWS CloudTrail when you start the session.

Enter reason

This value can have up to 256 characters.

Target instances

Filter instances

Instance ...	Instance ID	Agent ver...	Instance state	Availabili...	Platform	
+	Ubuntu22...	i-Of154d6...	3.3.2299.0	running	ap-south-1b	Ubuntu

Start session

Cancel

Next

Step 4:

Go inside Patch Manager. Click on “Start with an overview”, below “Create patch policy” option.

AWS Systems Manager

Review node insights

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Just-in-time node access [New](#)

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Automation

Change Calendar

Change Manager

Management & Governance

AWS Systems Manager Patch Manager

Manage patch compliance across the organization

Using Patch Manager, you can deploy patches simultaneously to applications and nodes across your organization. You can monitor patch compliance account by account.

Patch your instances

Expedite patching by creating a patch policy to apply operating system patches across the organization, and track compliance account by account.

Create patch policy

[Start with an overview](#)

How it works

Create patch policy

Specify in a policy whether to scan only or to install missing patches.

View dashboard

Get a high-level view of patch compliance for your applications and nodes.

View compliance reports

Examine the details of the resources that are not

Use cases and blog posts

[Learn more](#)

[Latest blog post](#)

More resources

[Documentation](#)

Click on Patch now.

Patch Manager [Info](#)[Patch now](#)[Create patch policy](#)

► Overview of patching operations - new

[Dashboard](#)[Compliance reporting](#)[Patch baselines](#)[Patches](#)[Patch groups](#)[Settings](#)

Node patching details (1)

[View log](#)[View detail](#)[Export to S3](#)[View all S3 exports](#)

1



	Name	Node ID	Patch configuration name	Patch configuration type	Compliance status	Critical non-compliant count
	Ubuntu22.04	i-0f154d67d5a0cf886	-	Patch group	Compliant	0

Use below configuration and then final click on Patch now. The Patch will start happening, you can see in Patch Manager Dashboard. Also the commands ran, any error came you can see in Run command option

Patch instances now [Info](#)

Basic configuration

Scan for missing patches or install patches, with or without rebooting. For more patching options, use the [Configure patching](#) page.

Patching operation

- ☒ Scan
- ☐ Scan and install

Instances to patch

Choose whether to patch all instances or only the instances you specify

- ☐ Patch all instances
- ☒ Patch only the target instances I specify

Target selection

Choose a method for selecting targets.

- ☐ Specify instance tags
- Specify one or more tag key-value pairs to select instances that share those tags.

- ☒ Choose instances manually
- Manually select the instances you want to register as targets.

- ☐ Choose a resource group
- Choose a resource group that includes the resources you want to target.

Managed instances (1/1)

Select one or more instances you want to patch.



1



☒	Name	Instance ID	Platform type	Operating system	Instance state	Agent version
☒	Ubuntu22.04	i-0f154d67d5a0cf886	Linux	Ubuntu	running	3.3.3598.0

Patching log storage

Select or create an S3 bucket for storing patching operation logs. Select **Do not store logs** if you don't require log information.

 [Patch now](#)

Explore nodes

Diagnose and remediate

Just-in-time node access New

Settings

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Automation

Change Calendar

[AWS Systems Manager](#) > Run Command

Commands

Command history

Command history

↺

View details

Rerun

Copy to new

Run command

Q

Search complete command history

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	Command ID	Status	Requested date	Document name	Comment
○	1333cbb2-d5b6-48d1-8335-4d88e611b5c6	✔ Success	Fri, 09 Jan 2026 10:39:29 GMT	AWS-UpdateSSMAgent	5a952668-5a25-412b-b999-97
○	68a8e56d-7596-4f6c-b539-06d25203ec7e	✔ Success	Fri, 09 Jan 2026 10:39:29 GMT	AWS-RefreshAssociation	Apply association with id at crea
○	b2ca844b-1461-4396-9db5-9339b6970386	✔ Success	Fri, 09 Jan 2026 10:32:21 GMT	AWS-RunPatchBaseline	0a4e24b5-e3b5-4cec-9038-a84
○	61dfacd0-f1d0-46fd-8dc4-b86efd07eccc	✔ Success	Fri, 09 Jan 2026 10:30:55 GMT	AWSFleetManager-GetFileSystemContent	
○	8e0f209b-dff2-4af1-bb71-9f6a9e72e6d6	✔ Success	Fri, 09 Jan 2026 10:30:48 GMT	AWSFleetManager-GetFileSystemContent	
○	b8117aef-2eb0-4406-8dd7-f5e80f8ac14c	✔ Success	Fri, 09 Jan 2026 10:30:39 GMT	AWSFleetManager-GetFileSystemContent	
○	2f628715-d595-4cb3-a42f-260b74b765c8	✔ Success	Fri, 09 Jan 2026 10:28:15 GMT	AWSFleetManager-GetFileSystemContent	

Then after 10-15 mins your compliance status will become Compliant.

[AWS Systems Manager](#) > [Patch Manager](#) > Compliance reporting

Patch Manager

Info

Patch now

Create patch policy ↗

► Overview of patching operations - new

Dashboard

Compliance reporting

Patch baselines

Patches

Patch groups

Settings

Node patching details (1)

View log ↗

View detail

Export to S3

View all S3 exports

Q

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⚙

	Name	Node ID	Patch configurati on name	Patch configurati on type	Compliance status	Critical non- compliant count
●	Ubuntu22.04	i-0f154d67d5a0cf886 ↗	-	Patch group	✔ Compliant	0

Now the Manual patch is over.